C-GR.2.3 · Weekly check · 5 questions

MA.7.GR.2.3 · Volume of a cylinder · Unit 12 Surface Area and Volume · Given Thursday, on the week's benchmark

Form title C-GR.2.3 Check · Volume of a cylinder
Description Learning goal: I can find the volume of a right circular cylinder. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

Short-answer validation (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

#	Question image	File · type	Options to add	Answer key	Pts
1	A cylindrical tank has a radius of 8 feet and a height of 12 feet. What is the volume of the tank? Round your answer to the nearest hundredth. Use 3.14 for π. Type a number only. No units, $ or %.	**Q1.png** Short answer	none (typed answer)	**2411.52** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
2	A cylindrical container has a diameter of 16 inches and a height of 24 inches. What is the volume of the container? Write your answer in terms of π. Type only the number in front of π. Example: for 7π, type 7.	**Q2.png** Short answer	none (typed answer)	**1536** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
3	A right circular cylinder has a radius of 6 units and a height of 16 units. What is the volume of the cylinder, in cubic units? A. 192π B. 1152π C. 576π D. 96π	**Q3.png** Multiple choice	A B C D	**C**	1
4	A cylindrical trash can has a diameter of 20 inches and a height of 26 inches. What is the volume of the trash can? Round your answer to the nearest hundredth. Use 3.14 for π. Type a number only. No units, $ or %.	**Q4.png** Short answer	none (typed answer)	**8164 · 8164.0 · 8164.00** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
5	Volume of a right circular cylinder: $V = \pi r^2 h$ Tank A is a cylinder with a radius of 4 feet and a height of 10 feet. Tank B is a cylinder with a diameter of 6 feet and a height of 15 feet. Which tank holds more, and by how many cubic feet? A. Tank B, by 380π ft³ B. Tank A, by 25π ft³ C. Tank B, by 25π ft³ D. Tank A, by 70π ft³	**Q5.png** Rigor Multiple choice Same problem, now multiple choice (paper: a written answer)	A B C D	**B**	1
6	LEARNING GOAL I can find the volume of a right circular cylinder. 4 I can solve a real-world volume problem, including comparing two cylinders. 3 I can find the volume of a cylinder from its radius or diameter and height. 2 I can find the area of the base. 1 I need help to start this.	**Level scale.png** Multiple choice (the template's last question)	Level 1 Level 2 Level 3 Level 4	no key	0

Answers with worked solutions are in the Answers folder next to the question images (same file names).